

AI Landslide Early Warning & Disaster Safety System

Aam Janta Aur Prashasan Ke Liye Pura Project Guide (Simple & Easy to Understand)

1. Yeh Project Kya Hai Aur Iska Maqsad Kya Hai? (Overview)

Pahadi ilaqa (jaise **Wayanad, Kerala, Shillong, Guwahati, Munnar, Aizawl**) me bhari baarish ke dauran achanak **bhuskhalan (landslide)** aane se hazaron logo ki jaan chali jaati hai aur raste band ho jaate hain. Yeh system **AI (Artificial Intelligence), NASA Satellite Data aur Live Mausam Vibhag (Weather API)** ki madad se bhuskhalan aane se **kuch ghante pehle hi logo ko alert kar deta hai** aur unhe bachne ka sabse surakshit rasta batata hai.

■ Real-Time Emergency Siren

Jaise hi khatra badhta hai (RED Alert), phone me automatic emergency siren bajta hai aur push notification aata hai.

■ Safe Road Detour Routing

Agar mukhya highway (NH-766) par malba gir gaya hai, toh system turant surakshit bypass (SH-59) ka rasta dikhata hai.

■ AI Photo Scanner

Raste me daraar ya mitti fisalne ki photo upload karo, AI turant scan karke khatre ki gambhirta bata deta hai.

■ Multilingual Voice Alerts

Padhe-likhe na hone par bhi app Hindi, English ya Assamese me bolkar alert sunata hai aur bolkar report likh sakte hain.

2. Citizen Portal: Aam Nagrik Ke Liye Kaun-Kaun Se Features Hain?

Citizen Portal aam janta ke liye banaya gaya hai jahan **bina kisi login ke** sirf ek click me apne ilaqa ki suraksha check ki ja sakti hai:

■ **GPS Auto-Zone Detection:** Aap jahan khade hain, app automatic aapki location detect karke sabse paas wale pahad ka live risk level bata deta hai.

■ 3-Color Live Risk Meter:

- **GREEN (Safe):** Mausam normal hai, ghabrane ki koi zaroorat nahi hai.
- **AMBER (Pre-Warning):** Bhari baarish ho rahi hai, alert rahein aur tayyari rakhein.
- **RED (Evacuate Now):** Mitti fisalne ka poora khatra hai, turant surakshit jagah par niklein!

■ **Live Mausam & Nami (Telemetry):** Pichle 24 ghante me kitni baarish hui aur mitti kitni geeli ho chuki hai (Topsoil Moisture), sab live dikhta hai.

■ **Bachav Ke Niyam (Survival Guide):** Aapatkaal ke waqt kya karna chahiye aur kin helplines (1070 / 1077) par phone karna hai, uski पूरी जानकारी.

3. 5 Naye Advanced Features (Jo Is Project Ko Sabse Alag Banate Hain)

Hackathon aur real disaster management ke liye humne isme **5 powerful features** add kiye hain:

A. ■ AI Computer Vision Hazard Scanner — Photo Se Daraar & Malba Pehchanne Ka AI

- Citizen jab sadak par darar ya mitti girne ki photo upload karta hai, AI image ko scan karta hai.
- Photo ke upar Red Bounding Box ban kar aa jata hai: **'Tension Crack (92.4% Confidence)'**.
- Report form me saari details (Category aur description) automatic bhara jaati hain.

B. ■■ 3D Pahad & Malba Runoff Simulation (Three.js) — Pahadi Dhalan Ka 3D Model

- NASA Satellite Radar ke data se pahad ka 3D model screen par ghumta hai.
- Steep slope angle (38.5 degree) ko 3D gradient me dikhata hai.
- Screen par malba (debris flow) girne ka animated rasta live dikhta hai taaki log samajh sakein ki malba kidhar aayega.

C. ■ Relief Camps & Bistar / Khane Ki Live Jaankari — Surakshit Shelter Resource Tracker

- Aas-paas ke sarkari relief camps ki list dikhata hai.
- **Bed Occupancy:** Kitne bistar khali hain aur kitne bhare hain (e.g. 215 / 350 beds occupied).
- **Rations:** Kitne din ka khana bacha hai (e.g. 7 Days Food Reserve), peene ka paani aur Doctor team ka contact.

D. ■ Offline SOS Mesh (Bina Internet Ke Madad) — Mobile Network Down Hone Par Bhi Kaam Karta Hai

- Bhushkhalan ke waqt mobile tower tootne par **Bluetooth Low Energy (BLE) Mesh** activate hota hai.
- Ek mobile se dusre mobile tak bina internet ke SOS signal travel karta hai aur rescue team tak pahunchta hai.

E. ■■ Multilingual Voice Assistant (Bolkar Alert Sunna) — Bhashini AI / Vernacular Voice

- Khatra aane par app **Hindi, English aur Assamese me bolkar** chetavni sunata hai.
- Report likhte waqt type karne ki zaroorat nahi — microphone daba kar bolenge toh text automatic likha jayega.

4. Live Project Links (Kahan Test Karein?)

■ Live Citizen App	https://frontend-eta-rouge-44.vercel.app/citizen	Citizen Safety Portal, 3D simulation, siren, relief camps
■ AI Photo Scanner	https://frontend-eta-rouge-44.vercel.app/report	Photo scan karke aapatkaal report submit karna
■■ 3D GIS Command Map	https://frontend-eta-rouge-44.vercel.app/sih-dashboard	Prashasan ke liye full-screen disaster map
■ Android Mobile APK	EWS-Landslide-AI-Early-Warning.apk (4.5 MB)	Mobile me install karke offline test karne ke liye
■ Complete ZIP File	EWS-Landslide-Full-Codebase.zip (16.8 MB)	Partner / Mentor ko share karne ke liye pura source code